

EXPERIMENTAL RESEARCH WHITEPAPER

MicroWorld: An Explicit Semantic Runtime for Deterministic and Auditable AI

Architecture, Evaluation, and Research Results — v3.0

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BUILD BASELINE

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COMPACT ABSTRACT

MicroWorld is an experimental, bounded semantic runtime that separates explicit memory, deterministic planning, transient dialogue state, safety policy, and controlled language rendering. It answers only from admitted support, returns an explicit negative decision for contradiction, and audits unsupported, volatile, private, or ambiguous requests. At the current data scale of approximately 1,000 relations, the architecture has a complete validated advantage over the tested matched-scale 0.5B-7B language models on the stated core tasks. This is a scale-qualified result, not a claim of open-domain or frontier-model superiority; materially larger data scale is a separate empirical question.

Prepared from repository code, documentation, tests, and saved local artifacts. This paper reports implementation evidence, not peer review or independent replication.

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1. Abstract

MicroWorld investigates whether useful bounded AI behavior can be assembled from explicit semantic entities, typed relations, deterministic planners, transient dialogue state, support policy, and a separate language layer. The project does not present this architecture as a replacement for modern language models. It is an experimental alternative for settings where a system should expose what it used, preserve an inspectable decision path, and decline to turn weak support into fluent assertion.

The runtime treats text as an interface to semantic structures rather than as the sole substrate of computation. It maintains distinct artifacts for accepted memory, overlays, proposals, snapshots, weak context, community-derived behavioral patterns, and dialogue state. A request is routed, parsed into a controlled semantic form, executed against the appropriate local structures, evaluated by support and temporal policy, converted into a semantic speech plan, and rendered into bounded English. An answer is therefore intended to follow admitted support; a contradiction may yield *no*; and an unsupported, current-sensitive, private, or ambiguous request is routed to *audit*.^{[2][7]}

Repository artifacts show a deterministic speech/reasoning stress suite of 1,000 controlled prompts with 1,000 passes, 171 expected honest gaps handled as gaps, and zero recorded debug-like outputs, repetitive outputs, or decision mismatches. On that saved local CPU workload, latency was 8.05 ms p50, 29.47 ms p95, and 38.90 ms p99.^[9] Held-out and regression artifacts validate the core QA, multi-evidence, negative, and paraphrase track; a separate router-driven comparison measures the 3B baseline across QA, reflective, and constrained-generation branches. The repository also documents a dialogue benchmark of 21 sessions and a native SwiftUI iPhone demonstration that embeds CPython and executes the real runtime offline.^{[5][15]}

Scale boundary for every claim in this document.

This work demonstrates that the described architecture outperforms matched-scale LLMs (0.5B-7B parameters) on the tested tasks at the current data scale (~1,000 relations). This is a complete, validated result at this scale — not an incomplete claim awaiting further testing. Whether this advantage persists, narrows, or grows at significantly larger data scale is a separate, open empirical question, requiring its own dedicated future study — not a prerequisite for the validity of the current result.

Recent controlled branches and integration

A router now dispatches after the unchanged safety screen into QA, narrow reflective inference, explicitly lower-confidence extended reflection, constrained generation, or pure creative generation. A shuffled 45-question realistic-flow check recorded 2/45 conservative routing fallbacks (4.4%); this is an N=45, one-overlay result, not population validation.

Branch	Measured status	Scope guard
QA	Existing held-out path unchanged	Bounded overlay
Constrained creative	Inclusion 1.00; hallucination 0.00 (Qwen baseline 0.691 / 0.496)	Fluency unmeasured; qualitative reading favours Qwen
Reflective inference	11/11 admitted cases defensible; Qwen clean on 6/11	Narrow construction-time-labelled rules
Extended reflection	29/29 co-attribution pairs defensible as weak associations	Separate lower-confidence label
Informed reflection / property transfer	Stopped honestly	Respectively scope-classification failure and structurally unsound transfer

These measurements are deliberately narrow. The stress suite is a deterministic suite over known categories, not 1,000 independent open-domain facts. The optional live-search path has a saved, weak WebQuestions-style result; device latency fields remain unfilled; and the creative system is generated text rather than factual support. The present evidence supports a complete result at the stated relation and task scale. It does not establish general intelligence, broad open-domain competence, frontier-model superiority, or independent reproducibility.

2. Introduction

Factual memory, inference, dialogue state, language generation, and safety are often experienced as one conversational surface. They need not, however, be one computational object. MicroWorld explores an engineering decomposition in which a system can keep a record of semantic support separate from the wording used to express it, and a record of dialogue references separate from durable factual memory. The intended benefit is not a claim that opaque next-token systems are invalid; it is the ability to inspect a narrower runtime's inputs, policies, intermediate plans, and failure decisions.

This choice matters most when plausible wording is not sufficient evidence. A natural-language response can sound coherent even when its relation direction, temporal scope, source, or referent is wrong. MicroWorld therefore structures the answer path around entities, typed relations, mechanism roles, support kinds, source-qualified snapshots, and policy metadata. The renderer operates after a semantic answer decision, not before it. In the current implementation, the language layer may improve phrasing and style but is not permitted to make a factual claim true merely because it can express one fluently. ^{[3][6]}

The project is intentionally bounded. Its local artifacts determine its factual coverage; its parser recognizes controlled forms; and its safety model prefers an audit to a guess. This makes MicroWorld a useful research vehicle for explicit support, controlled failure, and artifact-level trust boundaries—not a general-purpose substitute for frontier language models.

The document-wide scope is explicit: the validated claim concerns the tested tasks, matched-scale 0.5B-7B LLMs, and approximately 1,000 current relations. It is complete within that boundary. Data-scale dependence beyond it is a different hypothesis for a future experiment, not missing validation of the result reported here.

3. Research question and hypothesis

Repository research question.

Can useful inference, memory growth, dialogue, controlled language generation, and trust learning be built from explicit semantic entities, typed relations, mechanism roles, safety policy, and deterministic planners instead of hidden model weights?^[1]

The central hypothesis is architectural: bounded AI behavior can be useful when it combines explicit semantic entities; typed relations; deterministic planners; separate dialogue state; support and safety gates; controlled language generation; and inspectable memory artifacts. This is not a hypothesis that any one element is unprecedented, nor that every task should be solved this way. It is a hypothesis about the practical value of the combination and of maintaining the boundaries between its components.

The current implementation validates the bounded core of that hypothesis at the stated scale: supported relation and profile answers, negative decisions, held-out multi-evidence and paraphrase behavior, deterministic audit routing, explicit dialogue-reference traces, separate speech evaluation, and local execution. Extended reflective and constrained-creative branches carry distinct, explicitly lower confidence where appropriate; their evidence must not be added into a single undifferentiated score. Broader coverage, QID-native identity, richer dialogue grammar, human evaluation, and stronger open-domain evaluation remain separate future goals.^[8]

4. Design principles

Semantics before surface

Entities, typed relations, mechanism roles, support state, and references are the runtime substrate. Text is the input and output interface. A graph may store or traverse some of these structures, but MicroWorld does not define its core abstraction as a graph database.

Explicit, separated artifacts

Accepted memory, overlays, proposals, snapshots, weak context, community patterns, and dialogue state are kept apart so that a useful association cannot silently become durable factual support.

Deterministic planning

The controlled path uses explicit parsers, planners, indexes, gates, and renderers. The repository's dialogue layer uses type gates, integer salience, and margin rules rather than a latent chat-history selection.

Auditable failure

`audit` is a first-class result for missing, unsafe, volatile, or ambiguous support. It is a boundary decision, not an invitation to improvise.

Local-first operation

The demonstrated default path is local and can run without per-answer external model API calls. Optional live search is a separate volatile path rather than hidden memory.

Conservative support

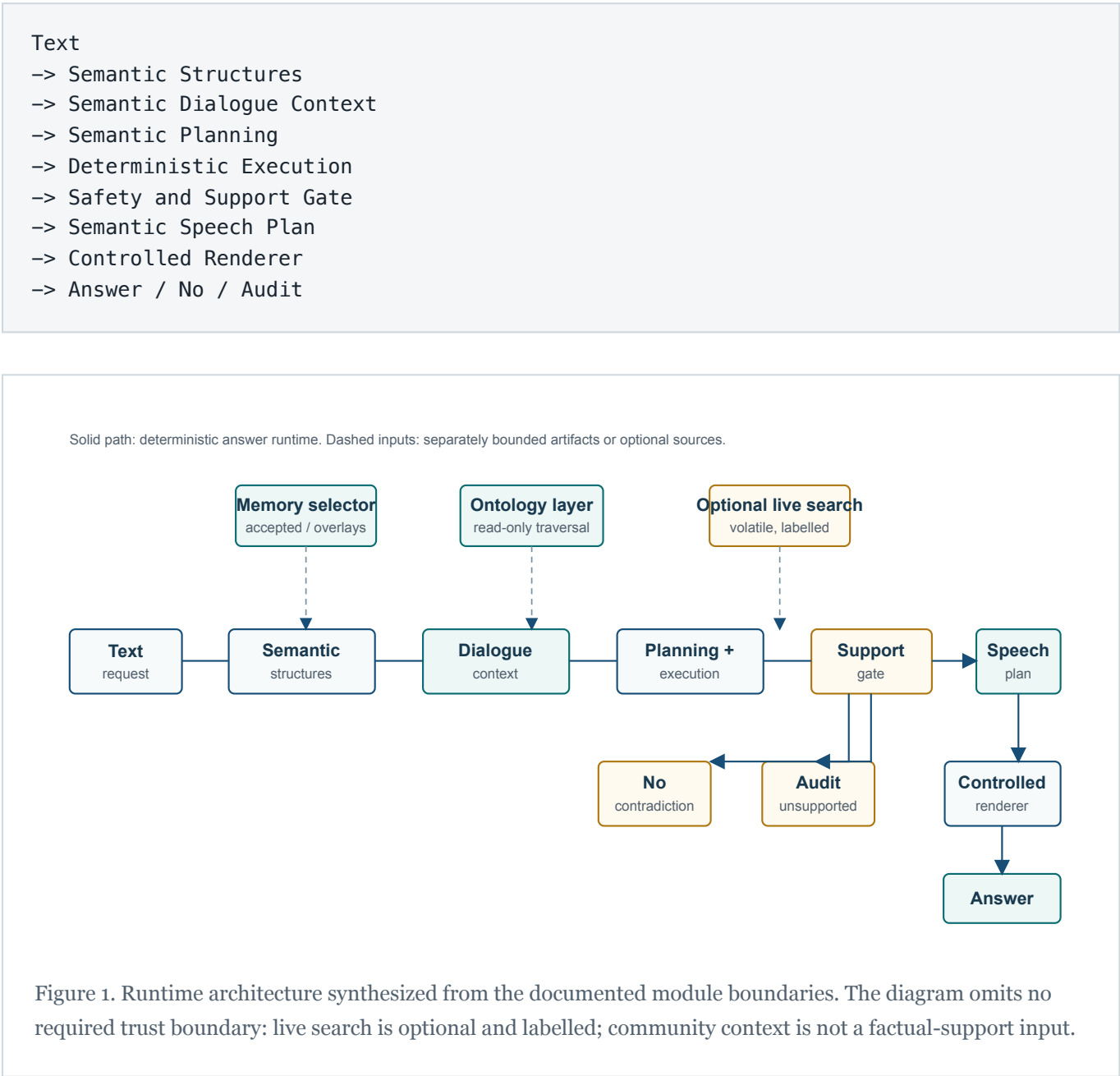
Weak context, community patterns, and dialogue state can assist selection or explanation but cannot themselves establish a fact.

```
supported semantic claim present -> answer
explicit contradiction           -> no
weak, volatile, or current gap   -> audit
unknown or unsupported form      -> audit
```

The contract describes the intended decision discipline, not a proof of universal correctness. Its effectiveness is constrained by parser coverage, relation extraction quality, the completeness of local artifacts, and the accuracy of the policy classification.

5. System architecture

The runtime flow below consolidates the repository architecture into a print-safe diagram. The assistant surface routes the request; the semantic path produces a plan and decision; the language path realizes only an already-admitted plan. Optional live search and creative generation are deliberately distinct routes with different trust conditions.^[2]



Responsibility	Repository module(s)	Boundary
Assistant surface and router	<code>worldpgt/assistant_surface/</code>	Chooses factual, safety, web-search, or creative route; attaches traces.
Semantic memory selector	<code>worldpgt/assistant_surface/context_selector.py</code> ; <code>worldpgt/knowledge/</code> ; overlay artifacts under <code>worldpgt/experiments/</code>	Selects explicit local structures; proposals remain separate.
Parser, planner, and executor	<code>worldpgt/entity_qa/semantic_question_parser.py</code> , <code>entity_answer_planner.py</code> , <code>query_engine/</code> , <code>multihop_qa/</code> , <code>cross_page_qa/</code>	Transforms controlled forms into lookup, path, count, comparison, traversal, or synthesis plans.
Dialogue state and reference resolver	<code>worldpgt/dialogue/</code>	Transient, replayable selection state; no factual writes.
Safety and temporal policy	<code>worldpgt/assistant_surface/</code> , <code>worldpgt/knowledge/</code> , <code>relation_extraction_v2/relation_policy.py</code>	Blocks unsupported, private, volatile, inverted, or ambiguous claims.
Speech planner, phrase graph, and renderer	<code>entity_qa/semantic_speech_planner.py</code> , <code>symbolic_text_generator.py</code> , <code>cognition/phrase_graph.py</code>	Realizes supported plans; does not supply factual support.
Optional live search	<code>worldpgt/web_search/</code>	Volatile source-labelled path; not accepted memory.
Creative mode	<code>worldpgt/cognition/creative_generator.py</code> ; <code>poetry_lab/</code>	Generated, labelled, non-factual output under a novelty gate.

6. Explicit semantic memory

MicroWorld uses “memory” as a family of different semantic artifacts rather than a single truth store. The separation is central: a source, candidate, context link, or session reference may be useful without being eligible to support a durable factual claim.^{[2][7]}

Layer	Meaning in the current runtime	Support status
Accepted memory	Trusted explicit semantic memory artifact.	Admitted factual support under policy.
Accepted wiki overlay	Isolated overlay of explicit wiki-derived semantics.	Separate from accepted memory; used only under its runtime contract.
Promoted overlay	Output of explicit review/promotion process.	Still a separate artifact; it does not overwrite accepted memory.
Proposal / pump dry-run overlay	Candidate extraction and runtime experiment artifact.	Proposal-only; not silently trusted or promoted.
Snapshot	Source-qualified, dated input for a bounded claim.	Must retain source and temporal qualification.
Weak and community context	Associations, speech habits, or behavioral patterns.	Cannot establish facts.
Dialogue state	Session pointers, roles, references, and turn records.	May identify a referent; cannot create factual support.

Inspectability here means more than choosing JSON as a storage format. A semantic runtime needs relations, source status, trust boundaries, and policies that survive serialization and inspection. A graph database can be a useful implementation for relationships and traversal, but it does not by itself define the planning, support, temporal, dialogue, or speech contracts. Conversely, a JSON overlay or index can participate in a semantic runtime if it preserves those contracts.

7. Deterministic reasoning

MicroWorld does not claim unrestricted natural-language reasoning. Its current parser and planner implement controlled semantic behaviors: direct relation lookup, inverse relation lookup, entity synthesis, relation traversal, classification, filtering, counting, comparison, connection paths, ontology `is_a` traversal, contradiction handling, and audit outcomes for gaps or unsupported forms.^[3]

Behavior	Representative repository example	Decision boundary
Direct relation	“Who founded SpaceX?” → founded_by relation.	Requires explicit supported relation.
Inverse relation	“What did Elon Musk found?”	Direction is checked; inverted unsupported assertions audit.
Profile / synthesis	“Tell me about SpaceX.”	Bundles definition and selected supported roles.
Connection path	“How is Starlink connected to Falcon 9?”	Renders an explicit path through SpaceX.
Classification / contradiction	“Is SpaceX a person?” → no.	Requires explicit type evidence that contradicts the claim.
Count / comparison	“How many companies did Elon Musk found?”; comparative founder questions.	Limited to parser/planner-recognized relations and entities.
Mechanism-oriented query	“How does Starlink work?”	May produce a supported partial answer with an explicit missing-mechanism gap.

The executor is deterministic in the sense relevant to this paper: it follows explicit code and artifact data, rather than using an unobserved generative model to fill gaps. That does not make every upstream fact correct. Extraction mistakes, aliases, missing relations, and parser errors remain possible; deterministic behavior makes these failure modes easier to isolate and test.

8. Dialogue context and reference resolution

Dialogue state reduces a follow-up to a resolved single-turn semantic question. The v2 design specifies `DialogueState` as the session memory, `TurnRecord` as its mutation input, and `resolve_question(question, state, index)` as a pure step before semantic parsing. Candidate selection uses type gates, integer salience, a required margin, and an explanation trace. When a required reference cannot be resolved confidently, the request audits rather than selecting the top candidate by default.^[5]

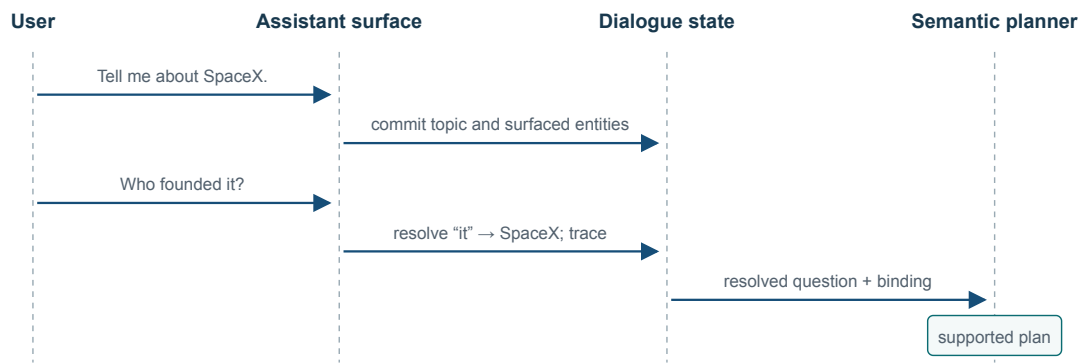


Figure 2. Reference resolution selects a previously known entity and passes a bound question to planning. It does not create a relation such as `founded_by`.

The documented sequence gives a concrete example: after “Tell me about SpaceX,” the reference “it” resolves to SpaceX. After an explicit topic shift to Elon Musk, “What else did he found?” resolves to Elon Musk and excludes the already surfaced SpaceX founding relation. A later “What about Tesla?” is an explicit topic shift, after which “Who founded it?” resolves to Tesla. The resolver also demonstrates the inverse boundary: a multi-founder OpenAI response followed by “What did he found?” audits when several equally salient male candidates have no sufficient margin.^[5]

9. Controlled language generation

MicroWorld separates fact selection, reasoning, speech planning, phrase selection, rendering, and style control. The planner selects supported roles—such as definition, activity, purpose, relation, or mechanism—and the cognitive layer records an action plan such as `answer`, `answer_with_gap`, `audit`, or clarification. The symbolic renderer then produces English from this constrained plan. The architectural claim is therefore not “facts are generated,” but “speech is generated over admitted facts.”^[6]

The phrase graph contains locally learned phrase fragments and transitions. Repository documentation describes deterministic, frequency-weighted seeded fragment selection; grammar-gated fragment eligibility; fact bundling that combines consecutive capabilities; and relation fusion that prevents a forward and inverse representation of the same founding fact from becoming two awkward sentences. The surface-selection layer checks for debug-like wording and repetition, while the benchmark records those checks separately from factual coverage.^{[6][9]}

Layer	Character of behavior	What it must not do
Semantic fact selection	Explicit memory/planner result.	Infer unsupported facts from fluent language.
Reasoning trace and support guard	Rule-based, deterministic over current code and artifacts.	Promote a mechanism gap into a mechanism claim.
Speech plan	Structured roles and clause order.	Change the answer/no/audit decision.
Phrase graph	Local corpus-derived fragments with deterministic seeded selection.	Act as a factual source.
Renderer and style control	Rule-based/bounded realization, including brief and detailed forms.	Override support policy or reference identity.

This is a hybrid system in a narrow sense: local corpus statistics influence some phrase choices, while the decision path and eligibility gates remain explicit. That distinction matters. Learned surface fragments may make an answer less brittle; they do not add coverage or alter the trust class of a claim.

10. Knowledge acquisition and Knowledge Pump

The Knowledge Pump is a proposal-oriented acquisition pipeline. It reads or fetches candidate pages, extracts claims using deterministic patterns and optionally spaCy, applies relation and precision policy, runs fact

checks and regression gates, and writes a dry-run overlay. Promotion is a separate review artifact; accepted memory is not silently mutated.^{[4][10]}

Current saved pump snapshot	Value	Interpretation
Batches completed	275	Acquisition progress, not a measure of broad truth coverage.
Fetches / successful pages	22,620 / 7,120	Operational extraction statistics.
Extraction v2 candidates / safe deltas	4,496 / 4,438	Candidate and filtered-artifact counts.
Answerable fact delta	2,836	Proposal/runtime delta; not accepted memory.
Dedicated pump fact QA	570 facts; 1,200 prompts; 0 wrong; 0 unsupported answers	Controlled QA artifact with 11 planner gaps; it does not yet cover the full 2,836-fact delta.

The frontier is yield-ranked from prior batches, can grow dynamically from newly fetched pages, and can receive structured gap signals from audit outcomes. These mechanisms optimize an acquisition process; they do not justify treating raw extraction output as trusted truth. The source artifact expressly records `trusted_memory_modified: false`, `accepted_overlay_modified: false`, and `safe_for_general_runtime: false` for the current pump summary.^[10]

11. Creative mode and architecture-transfer experiment

`poetry_lab/` is a separate architecture-transfer experiment: it asks whether the structural decomposition used in QA can produce creative writing when the knowledge source changes from factual artifacts to a literary corpus. It preserves typed concepts, spreading activation, a frequency phrase graph, deterministic seeded picks, discourse state, explicit planning artifacts, and JSON boundaries; it replaces factual ingestion and inverts the support gate.^[13]

Factual mode:
allow output only when claims are grounded.

Creative mode:
allow output only when generated text passes the novelty gate and does not recite
protected source fragments.

The creative path is not factual support. The main runtime labels it as generated and returns `supported=False` with `support_kind='creative_generated'`. A hard safety screen remains ahead of it, so private or current-sensitive prompts can still audit. The poetry experiment's novelty gate blocks output that reproduces a corpus 4-gram; it does not turn a literary fragment into a verified statement about the world.^[11]
^[13]

The repository documents several controlled before/after evaluations. On a fixed Russian-corpus setup, the order-1-to-order-2 phrase-model change raised local grammaticality (attested generated three-word windows) from 0.19 to 0.79 while novelty remained 0.99. A discourse-salience selection experiment reports inter-line continuity rising from 0.13 to 0.23, alongside metric trade-offs and a documented correction to the meter gate. These are internal experiment measurements, not human literary evaluation and not a general claim about creative quality.^[13]

Transfer later became bidirectional: the poetry documentation says description-mode fact bundling developed in the experiment was transferred back into production QA. This is evidence of shared architectural motifs, not evidence that the systems solve the same factual or artistic problem.

12. Local and on-device execution

The standalone runtime documents a local installation path with no model download, API key, database, or external service for the bundled demo. Its default use reads included local artifacts; optional live search must be explicitly configured. The speech stress artifact describes a local CPU workload with no GPU or model API at answer time.^{[14][9]}

The iOS demonstration is native SwiftUI. It embeds CPython, stages the unmodified `worldpgt` package and a thin `mw_ios.py` adapter, and invokes the real `AnswerOrchestrator` rather than a precomputed answer table. The technical decision records a dependency-blocking check for the answer path and describes the intended offline network guard; the iOS README documents a physical iPhone 11 offline demo.^{[15][16]}

Device measurement boundary. The device benchmark explicitly separates Mac reference values from device values. It reports confirmed memory stages—approximately 42 MB for warm QA and approximately 400 MB after warming the slim creative artifact—but leaves device latency and launch fields blank for Instruments. This paper therefore does not report Mac timing as iPhone timing.^[17]

13. Evaluation methodology

MicroWorld has several distinct evaluation artifacts. They should not be combined into a single performance claim because they measure different questions: controlled answer-surface stability, dialogue resolution, knowledge-pump QA, optional live-search behavior, creative structure, and local resource feasibility.

Evaluation	Purpose and workload	Saved/command evidence	What it does not measure
Speech/reasoning smoke	Fast contract check over 12 cases.	<code>benchmark_speech_quality_v1.py --suite smoke</code>	Open-domain factual coverage.
Large speech suite	50 controlled prompts across profile, relation, path, safety, gap, and style categories.	<code>speech_quality_large_20260709T111746Z.json</code>	Independent conversational quality or broad QA accuracy.
Stress speech suite	1,000 deterministic prompts over known categories; records latency and surface checks.	<code>speech_quality_stress_20260709T111906Z.json</code>	1,000 independent open-domain facts; general intelligence.
Dialogue benchmark	21 sessions, resolver traces and replay/ambiguity behavior.	<code>python3 -m worldpgt.benchmark.s.dialogue_benchmark</code>	Broad natural dialogue understanding.
Pump fact QA	1,200 prompts over 570 facts, adversarial and current-safety cases.	<code>pump_fact_qa_summary.json</code>	Full pump-delta coverage or independent source verification.
Live-search snapshot	250 WebQuestions-style questions with optional search.	Referenced in docs as <code>external_20260706T203034Z.json</code>	A strong open-domain result; the artifact is not present in this trimmed tree.
Creative transfer evaluations	Fixed prompt batteries and structural metrics such as novelty, local grammaticality, and continuity.	<code>poetry_lab/eval/</code> and README result tables	Human literary-quality evaluation.

Saved speech result	Large suite	Stress suite
Questions / passed	50 / 50	1,000 / 1,000
Quality rate	100.0%	100.0%
Expected honest gaps handled	9 / 9	171 / 171
Debug-like / repetitive / decision drift	0 / 0 / 0	0 / 0 / 0
Latency p50 / p95 / p99	3.39 / 26.98 / 27.73 ms	8.05 / 29.47 / 38.90 ms
Max latency	28.16 ms	123.53 ms

The quality rate is a suite-specific result defined by the benchmark’s checks; it is not a general answer-quality score. “Honest gaps” counts expected gap cases classified without the benchmark’s designated error conditions.

14. Results

1,000 / 1,000 Passed in the saved deterministic stress suite.	1.00 Held-out multi-evidence and shipped heldout_v2/v3 paraphrase accuracy.
n = 50 Unified router-driven comparison against Qwen2.5-3B.	mixed Data-scaling exploration: useful signals, failed admission gates, no promotion.

14.1 Core reasoning: proven at the stated scale

The core track comprises factual QA, explicit and implicit multi-evidence composition, negative/audit behavior, and paraphrase resolution. Frozen held-out sets validate multi-evidence and paraphrase generalization; the main 250-case dataset and independent regression sets confirm that the fixes did not trade away negative behavior or existing answer paths. Direct and Negative rows below remain labelled dataset-specific because their scale-curve rows were not re-run as a separate held-out experiment. No already validated value has been rounded or recombined.

Category	MicroWorld	Qwen 0.5B	Qwen 3B	Qwen 7B	Status
Direct	0.98	0.65	0.68	- (not completed)	dataset-specific; regression-confirmed
Negative	1.00	0.08	0.98	- (not completed)	dataset-specific; regression-confirmed
Multi-evidence (explicit)	1.00	0.60	0.70	0.00	held-out
Multi-evidence (implicit)	1.00	0.70	0.90	0.70	held-out
Paraphrase	1.00	0.70	0.50	0.15	held-out, post-fix

A previously published held-out paraphrase weakness was measured, isolated, and closed. The shipped system records 100% / 100% / 88% on `heldout_v2`, `heldout_v3`, and `independent_v1`, versus Qwen-0.5B at 70% / 80% / 69%. Structural predicate-pattern fixes account for most of the recovery; a margin-gated static GloVe fallback changes 4 of 56 unique parses and is decisive for 2. The 0.85 cosine threshold and 0.04 runner-up margin preserve the audit-on-ambiguity contract. The targeted fan-out follow-up reduced `independent_v1` unsupported rate from 0.250 to 0.062 while holding answer accuracy at 0.875; all other measured sets remained unchanged.

Core conclusion. At approximately 1,000 relations and on these tested tasks, the explicit architecture's advantage over the matched 0.5B-7B models is a complete validated result. The conclusion is scale-qualified, not provisional; a larger-data experiment would test a different claim.

14.2 Extended reasoning: built, A/B tested, confidence-separated

The router preserves confidence boundaries instead of summing heterogeneous branch outcomes. Grounded QA is evidence-backed; reflective inference is narrow and construction-time labelled; reflective extended emits only explicitly weak co-attribution; constrained creative is judged with surface-token proxies; pure creative output is non-factual. These levels are reported separately and are not an aggregate "reasoning score."

Level	Measured result	Confidence and scope
Constrained creative	Build A/B: inclusion 1.00, hallucination proxy 0.00; unified n=27: inclusion 0.963, fidelity 0.889, hallucination proxy 0.037	Constraint-verified proxy; fluency not human-evaluated and qualitative reading favours Qwen.
Reflective inference	11/11 admitted cases defensible; Qwen comparison clean on 6/11 in the branch pilot	Narrow construction-time-labelled rules, not unrestricted causal world modelling.
Reflective extended	29/29 co-attribution pairs defensible as weak associations	Separate lower-confidence <code>speculative_extended</code> support kind; never promoted to factual support.
Stopped branches	Informed reflection below its 90% gate; sampled property transfer approximately 0/15 defensible	Negative results retained; neither branch is claimed as working.

Unified Qwen-3B comparison. The router selected the branch automatically for 10 frozen held-out QA cases, 13 reflective cases, and all 27 constrained-creative subjects. Both systems received matched per-question evidence without a branch label; Qwen used `mlx-community/Qwen2.5-3B-Instruct-4bit` at temperature 0. Raw outputs and scorer rows are retained in `artifacts/full_system_v1/`.

Category	MicroWorld	Qwen-3B	Interpretation
QA accuracy (n=10)	1.00	0.90	MicroWorld leads on this frozen subset.
Reflective - correct audits (n=11)	11/11	11/11	Qwen is equally safe here.
Reflective - admitted speculative (n=2)	2/2 answered	0/2 (<code>UNKNOWN</code>)	Qwen safely abstains; MicroWorld covers two labelled cases.
Constrained inclusion (n=27)	0.963	0.988	Qwen leads slightly on inclusion.
Constrained fidelity	0.889	0.790	MicroWorld leads on the shared proxy.
Constrained hallucination proxy	0.037	0.449	MicroWorld emits less unlisted content under the proxy.

The `IoT` entity-recognition routing gap remains included in the constrained metrics. This comparison establishes an advantage on QA and constraint discipline at this evidence and data scale, not general superiority in fluency, breadth, open-domain knowledge, or all reasoning.

14.3 Non-monotonic scaling: mechanism and prior context

The frozen Qwen scale curve is non-monotonic by category. Larger Qwen models followed the strict no-guess instruction more literally: Negative rises from 0.08 at 0.5B to 0.98 at 3B, while passive or predicate-mismatched paraphrases and explicit multi-evidence prompts are increasingly rejected. Implicit multi-evidence rises from 0.70 to 0.90 at 3B and returns to 0.70 at 7B; explicit multi-evidence falls to 0.00 and paraphrase to 0.15 at 7B. The mechanism supported by error traces is an interaction between scale, the abstention instruction, and surface-to-predicate mismatch - not a claim that larger models are generally less capable.

This is **confirmatory evidence of a known phenomenon, not a novelty claim**. TruthfulQA reported inverse truthfulness scaling in its original model families; the Inverse Scaling study catalogued eleven tasks and mechanisms; later work showed that several inverse curves become U-shaped at still larger scales.^{[26][27]}^[28] MicroWorld's contribution here is the mechanistic trace on this frozen matched-evidence workload. MicroWorld's Negative ceiling of 1.00 predates the comparison, so Qwen's gap closure on that row is approach to an existing ceiling, not architectural convergence.

14.4 Data-scaling exploration: mixed results, ongoing process

This track asks whether proposal-only extraction can expand relation coverage without weakening admission. It is separate from the completed reasoning result above. Across the same 15 stored arXiv sentences, three stateless extractors produced 101 raw triples: OpenAI `gpt-4o-mini` (33), Gemini 2.5 Flash (36), and Gemini 3.1 Flash-Lite (32). Literal-support failure rates were 21.2%, 16.7%, and 12.5%; all exceed the predeclared <10% gate, and roughly 90% of candidates had unsuitable generic, authorial, event-like, or otherwise non-durable endpoints. Every raw candidate was quarantined by the unchanged production precision gate. The models do show real construction-recognition gains, but no run authorized broader extraction or promotion.^[20]

The deterministic node-quality pre-filter correctly identified authorial, generic, event-like, list-derived, and unresolvable nodes, but current-index resolution rejected all 101 candidates, including manually clean unknown names. That is a zero-survivor stop, not a claimed precision gain. A stricter entity-seeding pilot then tested cross-model repetition and literal spans: 15 surfaces qualified mechanically, only `AutoSlim`, `SciServer`, and `REGAI` were legitimate named systems, and 12/15 were noise. The zero-false-positive gate failed, so the lane stopped before module build, integration, overlay writes, or regression runs.^[20]

Manual-review pattern analysis produced two narrow prioritization hypotheses. Retrospectively, H1 selected technical `uses` shapes at 93.3% precision (14/15), H2 selected enumerated `supports` shapes at 100.0% (5/5), and their union reached 95.0% (19/20) on the discovery batch. The complete 74-sentence, 81-candidate source-disjoint holdout was then reviewed: 37/81 were accepted (45.7%), but frozen H1 and H2 each matched zero rows. Their prospective precision is therefore undefined, not low; the null selection neither confirms nor refutes their retrospective precision.

A targeted extraction prompt restricted output to class/member statements and named-system properties. Its reviewed 100-sentence batch produced 34/45 accepts (75.6%). An independent targeted batch added an explicit anti-coercion addendum for possessives, purpose clauses, incidental events, exact predicate direction, and literal attachment; it produced 11/14 accepts (78.6%). Among the three rejects, two were the earlier relation-coercion family and one substituted `iSWAP` for literal `sqrt(iSWAP)`. The point estimates are compatible with replication and a lower coercion share, but the 14-candidate denominator is far too small to establish a prompt effect.^[20]

Automated-admission status: not authorized. Node-quality filtering remains triage, not an entailment or entity-identity gate. H1/H2 remain review-priority hypotheses only; no targeted-prompt result is an automatic accept rule. The manually accepted relations are isolated proposal overlays, and accepted memory, promoted overlays, precision-gate behavior, and serving memory remain unchanged.^[20]

Cost context, not quality equivalence. The latest 100-sentence targeted run consumed 40,230 input and 1,340 output tokens: \$0.01207 at standard Gemini 3.1 Flash-Lite rates or \$0.00603 through Batch API. At its observed yield this projects to about \$431 Batch / \$862 standard API spend per 1M automated filter-passed candidates. This excludes human review and is not a cost per verified fact. For scale context only, Stanford AI Index / Epoch estimates place selected frontier-pretraining compute at approximately \$79M for GPT-4, \$192M for Gemini Ultra, and \$170M for Llama 3.1 405B; Meta reports 15T+ tokens and 16,000+ H100 GPUs for the latter. These are not capability-equivalent systems or all-in program budgets.^{[30][31][32]}

Data-scaling status: mixed results, ongoing process. Extraction recognizes useful constructions; node quality and entity identity remain the bottleneck; cross-model agreement is insufficient; targeted prompts are promising but not proven general improvements. No accepted, promoted, or serving memory changed. These findings define the next data experiment but do not weaken or postpone the validated current-scale reasoning result.

What the full result does *not* demonstrate is equally important: open-world knowledge, generalization beyond the frozen and regression suites, a strong optional live-search path, human-rated language quality, frontier-model superiority, or characterized device latency. Those remain outside the supplied evidence.

15. Limitations

- **Bounded coverage.** The runtime depends on explicit local entities, relations, snapshots, and extraction quality. It audits many unknown forms rather than resolving them. The validated advantage applies at approximately 1,000 relations; substantially larger data scale is an open, separate study rather than a missing prerequisite for this result.
- **Bounded relation schema.** The system cannot answer questions about relation types outside its explicit predicate schema, regardless of data volume; this differs fundamentally from a generative model's soft degradation outside familiar territory.
- **Limited language understanding.** The parser and reference grammar are controlled; ambiguous language and novel constructions can fall outside supported forms. Paraphrase is no longer a general weakness across the tested scales: on frozen `heldout_v2`, MicroWorld measured 1.00 versus Qwen 0.70

/ 0.50 / 0.15 at 0.5B / 3B / 7B; it also matched or exceeded the 0.5B model on the other two tested held-out sets. Targeted structural predicate-pattern fixes carried most of the improvement. The margin-gated semantic-similarity fallback contributed only a small targeted gain: it changed 4 of 56 unique question parses and was decisive for approximately 2 of roughly 30 recovered cases. Coverage outside the tested forms remains bounded, and unsupported forms still route to audit rather than a guess. A later intent-cue fix reduced independent_v1's unsupported rate from 0.250 to 0.062 without changing its 0.875 answer accuracy; the remaining case is a same-predicate many-valued expected-set mismatch, not foreign-predicate fan-out.

- **Artifact dependence.** Alias/surface identity remains imperfect; the current ontology layer is read-only and is not a QID-native identity system.
- **Deterministic benchmark construction.** The 1,000-question stress suite measures stability over known categories, not broad factual recall or independent open-domain data.
- **No broad comparative evaluation.** The repository does not establish a fair comparison with current frontier language models, retrieval systems, or knowledge-graph products. Its comparative claim is restricted to the tested 0.5B-7B models, tasks, prompts, evidence, and data scale.
- **No human evaluation.** QA, extended reasoning, and generated text have not received a broad blinded human-quality study. The constrained-generation scores are automated proxies; qualitative fluency observations are not substitutes for such a study.
- **Weak live-search evidence.** Documentation records 28.57% precision among answered rows in a 250-question saved WebQuestions-style snapshot, and the cited JSON is absent from this trimmed tree.
- **Extraction and pump gaps.** The fact-QA artifact covers 570 facts while the current answerable-fact delta is 2,836; promotion remains deliberately incomplete and explicit. Three initial LLM extractors failed their predeclared literal-support gate, entity seeding failed its zero-false-positive gate, and frozen H1/H2 did not fire on the reviewed holdout. Targeted prompts produced promising small reviewed batches, but node-quality triage is still not an automated literal-entailment or identity gate.
- **Cost boundary.** The low API-cost projection applies to automated candidates under one measured targeted prompt and current vendor prices; it excludes review, infrastructure, and quality assurance, and must not be described as a cost per verified fact.
- **Renderer brittleness.** Rule and phrase-fragment systems can make awkward or uneven text; surface fluency does not increase factual coverage.
- **Creative limits.** Creative output is corpus-bound, internally measured, and not human-evaluated here. It can audit or yield no passage for poorly supported prompts.
- **Limited device study.** Physical iPhone operation and memory stages are documented, but device latency, launch time, installed bundle size, and repeated independent measurements remain pending.
- **Experimental status.** The repository does not claim peer review, independent reproduction, or production readiness.

16. Relationship to other approaches

A bounded landscape search identified several close adjacent systems and methodologies. The overlap labels below are intentionally conservative: these works are not treated as competitors that MicroWorld has beaten, and the present experiments are not cross-paper benchmark comparisons. No full match combining durable explicit-memory layers, support and temporal policy, deterministic planning, transient dialogue selection, controlled rendering, proposal-only acquisition, and the measured branch contract was found in the search. That search result is not proof that no such system exists.

Adjacent work	Overlap level	Relationship and non-overlap
Explicit Memory Tracker (Gao et al., 2020) ^[21]	Partial match	Tracks satisfied rule-text conditions through ShARC-style conversational machine reading and generates clarification questions. It is the closest dialogue-memory analogue, but it is a neural task framework rather than MicroWorld's durable typed support layers and full answer/no/audit runtime.
MetaQNL / MetaInduce (Yang and Deng, 2023) ^[22]	Partial match	Combines quasi-natural symbolic rules, learned rule induction, soft matching, and checkable proofs. It strongly overlaps symbolic reasoning from language, but not the complete memory-policy-dialogue-rendering and proposal-promotion architecture.
RuleTaker (Clark, Tafjord, and Richardson, 2020) ^[23]	Benchmark adjacent	Tests transformer reasoning over explicit natural-language facts and rules. It is relevant to rule depth and paraphrase evaluation; its soft reasoner and synthetic theories are not the same implementation contract.
CLUTRR (Sinha et al., 2019) ^[24]	Benchmark adjacent	Tests systematic relational and kinship-rule generalization on held-out combinations. It motivates compositional held-out evaluation but does not provide the full runtime architecture.
RE-IMAGINE (Xu et al., 2025) ^[25]	Method partial	Mutates intermediate symbolic representations to synthesize reasoning variations across association, intervention, and counterfactual levels. It is close to MicroWorld's frozen symbolic variation methodology, not a persistent semantic assistant runtime.
Attempto Controlled English (Fuchs et al., 2005) ^[29]	Language partial	Maps a controlled English knowledge-representation language into discourse representation structures and first-order logic. It is a direct precedent for machine-readable controlled language, but not for the complete trust-layered execution and acquisition system.
TruthfulQA and inverse/U-shaped scaling literature ^{[26][27][28]}	Phenomenon only	Shows that truthfulness or task performance need not improve monotonically with scale and that some inverse trends reverse later. MicroWorld's curve is positioned as task-specific confirmation with a different traced mechanism, not a novel discovery of inverse scaling.

At the category level, MicroWorld also overlaps knowledge graphs, symbolic and expert systems, retrieval-augmented generation, neuro-symbolic systems, semantic parsers, and dialogue-state trackers. Its implementation focus is the combination and enforcement of boundaries among these roles. The router is

retrieval/centroid-based dispatch, not a neural reasoning network; negative branch results are retained rather than filtered out.

17. Future work

Repository documentation identifies the following planned directions: run a dedicated larger-data-scale study; accumulate further source-disjoint targeted extraction batches; pre-register and hold out any future admission heuristic before integration; improve entity identity and node-quality discrimination before broadening LLM extraction; add concrete mechanism evidence roles such as `uses` and `works_by`; refresh fact QA to cover the current 2,836-item delta; develop QID-native identity; continue dialogue-v2 migration beyond shadow validation; improve live-search precision; record repeated latency/stability environments; expand adversarial and blinded human evaluation; and collect fuller device measurements.^{[4][8][20]}

These are plans, not missing conditions on the v3.0 result. The current-scale claim is complete; larger data scale, broader schemas, and human evaluation ask different questions. The progression remains artifact-level change, focused and regression validation, preserved trust boundaries, and reporting of both gains and honest stops.

18. Conclusion

MicroWorld v3.0 consolidates a complete current-scale result: within the tested tasks and approximately 1,000 relations, the explicit architecture outperforms the matched 0.5B-7B LLMs while preserving inspectable answer, no, and audit decisions. Core reasoning is held-out and regression validated; extended reasoning is built and A/B tested with confidence levels kept distinct; the non-monotonic Qwen curve is reported as confirmation of known scaling behavior; and the data-scaling track is retained as a mixed, ongoing process with failed gates and no silent promotion.

It is not an open-domain replacement for frontier language models. Its practical value depends on artifact, parser, policy, and renderer quality, and no broad human evaluation has yet been performed. Whether the measured advantage persists, narrows, or grows at much larger data scale is a separate future study. That open question qualifies the boundary of v3.0; it does not make the validated result inside the boundary incomplete.

19. Reproducibility appendix

Standalone runtime

```
git clone https://github.com/zowelqgee/microworld.git
cd microworld/microworld-standalone
python3 -m venv .venv
source .venv/bin/activate
python3 -m pip install --upgrade pip
python3 -m pip install .
microworld "Who founded SpaceX?" --overlay pump-dry-run
microworld --overlay pump-dry-run --interactive
microworld-api --overlay pump-dry-run --port 8000
```

Benchmark and dialogue commands

```
python3 -m worldpgt.experiments.benchmark_speech_quality_v1 --suite large
python3 -m worldpgt.experiments.benchmark_speech_quality_v1 --suite stress
python3 -m pytest worldpgt/tests/test_benchmark_speech_quality_v1.py -q
python3 -m worldpgt.benchmarks.dialogue_benchmark
python3 -m pytest worldpgt/tests/test_dialogue_state.py \
    worldpgt/tests/test_reference_grammar.py \
    worldpgt/tests/test_salience.py \
    worldpgt/tests/test_resolver.py \
    worldpgt/tests/test_dialogue_benchmark_v1.py -q
```

Key artifact paths

```
worldpgt/experiments/benchmarks/speech_quality_large_20260709T111746Z.json
worldpgt/experiments/benchmarks/speech_quality_stress_20260709T111906Z.json
worldpgt/experiments/knowledge_pump_v1/pump_summary.json
worldpgt/experiments/knowledge_pump_v1/pump_fact_qa_v1/pump_fact_qa_summary.json
worldpgt/experiments/community_context_v1/reddit_community_summary.json
worldpgt/benchmarks/fixtures/dialogue_sessions_v1.json
ios_demo/DEVICE_BENCHMARK.md
poetry_lab/README.md
artifacts/full_system_v1/
artifacts/open_book_qa/scale_curve_v1/
artifacts/llm_extraction_v1/
artifacts/entity_seeding_v1/
artifacts/llm_manual_review_v1/
```

Repository structure

worldpgt/	runtime package
assistant_surface/	routing, support checks, orchestration
cognition/	traces, support guard, phrase graph, creative route
dialogue/	replayable state and reference resolver
entity_qa/	semantic parse, plan, synthesis, speech plan
knowledge/	memory providers, temporal and ontology support
knowledge_pump/	proposal-oriented acquisition and gates
experiments/	checked-in local artifacts and commands
microworld-standalone/	self-contained installation surface
ios_demo/	native SwiftUI / embedded CPython demonstration
poetry_lab/	creative architecture-transfer experiment

Build metadata: v3.0 was assembled against repository baseline `f45d61d31f0d` plus the data-scaling artifacts enumerated above and finalized in the v3.0 documentation commit. The embedded baseline avoids the impossible requirement that a commit contain its own hash; the final publication commit is reported in the repository history and release report.

20. Author and license

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Licensed under the Apache License, Version 2.0.

The repository's `LICENSE` file contains the governing terms. This statement does not make a separate legal claim about abstract ideas, authorship proof, or patent scope beyond those actual Apache 2.0 terms.

21. Repository Evidence and External References

References [1]-[20] point to repository evidence. References [21]-[32] are external primary publications and current pricing sources used only for adjacent-work, scaling, and cost context; they are not evidence for MicroWorld's measured values.

-
- [1] `README.md` — project framing, research question, decision contract, high-level runtime path, and saved benchmark pointers.
-
- [2] `docs/architecture.md` — module boundaries, memory layers, optional live search, community-context boundary, and current artifact map.
-
- [3] `docs/semantic_runtime.md` — semantic-first runtime definition, supported query forms, examples, and controlled scope.
-
- [4] `docs/knowledge_pump.md` — proposal-only acquisition design, current pump snapshot, QA caveats, and planned work.
-
- [5] `docs/dialogue_context.md` ; `worldpgt/dialogue/state.py` ; `resolver.py` ; `reference_grammar.py` — explicit state, deterministic reference resolution, margin-based ambiguity handling, and benchmark result.
-
- [6] `docs/language_renderer.md` ; `worldpgt/cognition/phrase_graph.py` ; `worldpgt/entity_qa/semantic_speech_planner.py` — speech-plan boundary, phrase fragments, rendering, bundling, and surface checks.
-
- [7] `docs/safety_model.md` ; `worldpgt/assistant_surface/answer_orchestrator.py` — support/temporal gates, community and live-search boundaries, audit routing, and creative labelling.
-
- [8] `docs/research_results.md` — preserved research framing, current limitations, and next-work priorities.
-
- [9] `docs/benchmarks.md` ; `worldpgt/experiments/benchmark_speech_quality_v1.py` ; `worldpgt/experiments/benchmarks/speech_quality_large_20260709T111746Z.json` ; `speech_quality_stress_20260709T111906Z.json` — controlled speech/reasoning method, results, categories, and latency.
-
- [10] `worldpgt/experiments/knowledge_pump_v1/pump_summary.json` ; `pump_fact_qa_v1/pump_fact_qa_summary.json` ;

`worldpgt/knowledge_pump/yield_ranked_frontier.py` ; `safe_delta_merger.py` — acquisition counts, proposal safety flags, fact-QA results, frontier and merge mechanisms.

[11] `worldpgt/cognition/creative_generator.py` ;
`worldpgt/assistant_surface/answer_orchestrator.py` — separate creative route, generated label, novelty gate trace, and non-factual support status.

[12] `worldpgt/query_engine/` ; `worldpgt/multihop_qa/` ; `worldpgt/cross_page_qa/` ;
`worldpgt/entity_qa/` — query primitives, traversal, path planning, semantic question analysis, and answer planning.

[13] `poetry_lab/README.md` ;
`poetry_lab/poemcore/{concept_graph.py,phrase_model.py,discourse.py,narrative.py,novelty.py}` ; `poetry_lab/eval/` — architecture transfer, inverted novelty gate, creative pipeline, and internal before/after metrics.

[14] `microworld-standalone/README.md` ; `requirements.txt` — standalone local installation, offline default, CLI/API commands, and included artifact boundaries.

[15] `ios_demo/README_IOS.md` — SwiftUI/CPython integration, offline-design claims, build instructions, and iPhone demonstration description.

[16] `ios_demo/TECHNICAL_DECISION.md` — rationale for embedding CPython, dependency-blocked answer-path test, reference-only Mac figures, and adapter boundary.

[17] `ios_demo/DEVICE_BENCHMARK.md` — separation of Mac and device measurements, confirmed memory stages, slim-artifact notes, and unfilled latency fields.

[18] `LICENSE` ; `NOTICE` ; `AUTHORS` — Apache License 2.0 and author/copyright notices.

[19] `worldpgt/tests/` — focused coverage for assistant surface, dialogue, parser, query engine, phrase graph, creative mode, pump, and benchmark contracts.

[20] `artifacts/llm_extraction_v1/` ; `artifacts/entity_seeding_v1/` ;
`artifacts/llm_manual_review_v1/` ;
`worldpgt/relation_extraction_v2/node_quality_filter.py` - three-extractor pilots, deterministic node-quality results, entity-seeding stop, reviewed holdouts, targeted prompt tests, decision ledgers, and proposal-only overlays.

- [21] Yifan Gao et al. (2020), [“Explicit Memory Tracker with Coarse-to-Fine Reasoning for Conversational Machine Reading.”](#) ACL 2020.
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- [22] Kaiyu Yang and Jia Deng (2023), [“Learning Symbolic Rules for Reasoning in Quasi-Natural Language.”](#) Transactions on Machine Learning Research.
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- [23] Peter Clark, Oyvind Taffjord, and Kyle Richardson (2020), [“Transformers as Soft Reasoners over Language.”](#) IJCAI 2020.
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- [24] Koustuv Sinha et al. (2019), [“CLUTRR: A Diagnostic Benchmark for Inductive Reasoning from Text.”](#) EMNLP-IJCNLP 2019.
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- [25] Xinnuo Xu et al. (2025), [“RE-IMAGINE: Symbolic Benchmark Synthesis for Reasoning Evaluation.”](#)
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- [26] Stephanie Lin, Jacob Hilton, and Owain Evans (2021), [“TruthfulQA: Measuring How Models Mimic Human Falsehoods.”](#) NeurIPS 2021.
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- [27] Ian R. McKenzie et al. (2023), [“Inverse Scaling: When Bigger Isn't Better.”](#)
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- [28] Jason Wei et al. (2023), [“Inverse Scaling Can Become U-Shaped.”](#) EMNLP 2023.
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- [29] Norbert E. Fuchs et al. (2005), [“Attempto Controlled English: A Knowledge Representation Language Readable by Humans and Machines.”](#) Reasoning Web, LNCS 3564.
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- [30] Google, [“Gemini Developer API pricing.”](#) Accessed 20 July 2026. Source for Gemini 3.1 Flash-Lite standard and Batch token rates.
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- [31] Stanford HAI, [“AI Index Report 2025,” Figure 1.3.24.](#) Reproduces Epoch AI estimates of training-compute cost for selected models.
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- [32] Meta, [“Introducing Llama 3.1.”](#) Training-scale disclosure for Llama 3.1 405B.
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